DM872

Mathematical Optimization at Work

Benders' Algorithm

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Outline

1. Dantzig Wolfe

2. Benders Decomposition

3. Benders Algorithm

Unboundedness

$$\begin{array}{rcrcl} \max & 2x_1 & + & x_2 & \\ & & & x_2 & \leq & 5 \\ & -x_1 & + & x_2 & \leq & 1 \\ & & & x_1, x_2 & \geq & 0 \end{array}$$

- Initial tableau

	x1	x2	x3	x4	-z	b
x3	0	1	1	0	0	5
x4	-1	1	0	1	0	1
	2	1	0	0	1	0

- x_2 entering, x_4 leaving

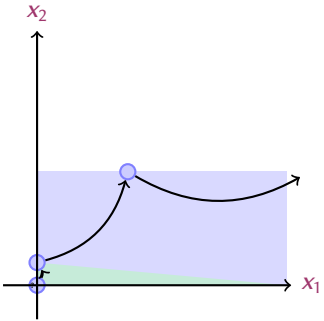
	x1	x2	x3	x4	-z	b
II'=II-I'	1	0	1	-1	0	4
I'=I	-1	1	0	1	0	1
III'=III-I'	3	0	0	-1	1	-1

$-x_1 + x_2 + x_4 = 1$, x_1 can increase without restriction, $\theta = \min\{\frac{b_i}{a_{is}} : a_{is} > 0, i = 1 \ldots, n\}$

- x_1 entering, x_3 leaving

	x1	x2	x3	x4	-z	b
I'=I	1	0	1	-1	0	4
II'=II+I'	0	1	1	0	0	5
III'=III-3I'	0	0	-3	2	1	-13

x_4 was already in basis but for both I and II ($x_2 + 0x_4 = 5$), x_4 can increase arbitrarily



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Farkas Lemma

Lemma (Farkas)

Let $A \in \mathbb{R}^{m \times n}$ *and* $\boldsymbol{b} \in \mathbb{R}^m$*. Then,*

$$\begin{aligned} \textit{either} \quad & \textit{I.} & \exists \boldsymbol{x} \in \mathbb{R}^n : A\boldsymbol{x} = \boldsymbol{b} \textit{ and } \boldsymbol{x} \geq \boldsymbol{0} \\ \textit{or} \quad & \textit{II.} & \exists \boldsymbol{y} \in \mathbb{R}^m : \boldsymbol{y}^T A \geq 0^T \textit{ and } \boldsymbol{y}^T \boldsymbol{b} < \boldsymbol{0} \end{aligned}$$

Easy to see that both I and II cannot occur together:

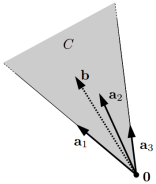
$$(0 \leq) \qquad \boldsymbol{y}^T A\boldsymbol{x} = \boldsymbol{y}^T \boldsymbol{b} \qquad (< 0)$$

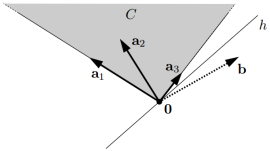
Geometric interpretation of Farkas L.

Linear combination of $\boldsymbol{a}_i$ with nonnegative terms generates a convex cone:

$$\{\lambda_1 \boldsymbol{a}_1 + \ldots + \lambda_n \boldsymbol{a}_n, \mid \lambda_1, \ldots, \lambda_n \geq \boldsymbol{0}\}$$

Polyhedral cone: $C = \{\boldsymbol{x} \mid A\boldsymbol{x} \leq \boldsymbol{0}\}$, intersection of many $\boldsymbol{a}\boldsymbol{x} \leq 0$
Convex hull of rays $\boldsymbol{p}_i = \{\lambda_i \boldsymbol{a}_i, \lambda_i \geq 0\}$





Either point $\boldsymbol{b}$ lies in convex cone C
or $\exists$ hyperplane h passing through point 0 $h = \{\boldsymbol{x} \in \mathbb{R}^m : \boldsymbol{y}^T \boldsymbol{x} = 0\}$ for $\boldsymbol{y} \in \mathbb{R}^m$ such that all vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ (and thus C) lie on one side and $\boldsymbol{b}$ lies (strictly) on the other side (ie, $\boldsymbol{y}^T \boldsymbol{a}_i \geq 0, \forall i = 1 \ldots n$ and $\boldsymbol{y}^T \boldsymbol{b} < 0$).

Variants of Farkas Lemma

Corollary

(i) $A\boldsymbol{x} = \boldsymbol{b}$ *has sol* $\boldsymbol{x} \geq \boldsymbol{0} \iff \forall \boldsymbol{y} \in \mathbb{R}^m$ *with* $\boldsymbol{y}^T A \geq \boldsymbol{0}^T, \boldsymbol{y}^T \boldsymbol{b} \geq \boldsymbol{0}$

(ii) $A\boldsymbol{x} \leq \boldsymbol{b}$ *has sol* $\boldsymbol{x} \geq \boldsymbol{0} \iff \forall \boldsymbol{y} \geq \boldsymbol{0}$ *with* $\boldsymbol{y}^T A \geq \boldsymbol{0}^T, \boldsymbol{y}^T \boldsymbol{b} \geq \boldsymbol{0}$

(iii) $A\boldsymbol{x} \leq \boldsymbol{0}$ *has sol* $\boldsymbol{x} \in \mathbb{R}^n \iff \forall \boldsymbol{y} \geq \boldsymbol{0}$ *with* $\boldsymbol{y}^T A = \boldsymbol{0}^T, \boldsymbol{y}^T \boldsymbol{b} \geq \boldsymbol{0}$

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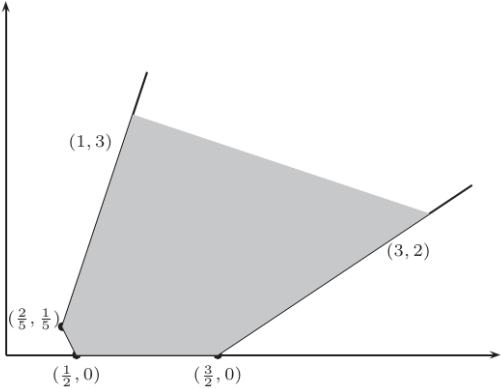
$$U = \{u \in \mathbb{R}^2_+ : 4u_1 + 2u_2 \geq 2, -2u_1 + 3u_2 \geq -3, 3u_1 - u_2 \geq 1\}$$

has extreme points $(\frac{2}{5}, \frac{1}{5})^T, (\frac{1}{2}, 0)^T, (\frac{3}{2}, 0)^T$ and extreme rays $(3, 2)^T, (1, 3)^T$ giving the extended formulation:

$$U = \{u \in \mathbb{R}^2 : u = \begin{pmatrix} \frac{2}{5} \\ \frac{1}{5} \end{pmatrix} \lambda_1 + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \lambda_2 + \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} \lambda_3 + \begin{pmatrix} 3 \\ 2 \end{pmatrix} \mu_1 + \begin{pmatrix} 1 \\ 3 \end{pmatrix} \mu_2,$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \ (\lambda, \mu) \in \mathbb{R}^3_+ \times \mathbb{R}^2_+\}. \qquad \square$$



Example 12.1 Consider the mixed integer program

$$\begin{array}{rrrrrrl}
\max & 4x_1 & +7x_2 & +2y_1 & -3y_2 & +y_3 & \\
 & 2x_1 & +4x_2 & +4y_1 & -2y_2 & +3y_3 & \leq 12 \\
 & 3x_1 & +5x_2 & +2y_1 & +3y_2 & -y_3 & \leq 10 \\
 & x & \leq 2, & x \in \mathbb{Z}^2_+, & y \in \mathbb{R}^3_+. & &
\end{array}$$

The corresponding dual polyhedron $U = \{u \in \mathbb{R}^m_+ : uG \geq h\}$ as well as its extreme points and extreme rays have been given in Example 2.4. So the complete Benders' reformulation (12.4)–(12.7) is as follows:

$$\begin{array}{rrrrrl}
\max & 4x_1 & +7x_2 & +\eta & & \\
 & 12x_1 & +22x_2 & & \leq & 56 \quad v^1 = (3,2) \\
 & 11x_1 & +19x_2 & & \leq & 42 \quad v^2 = (1,3) \\
 & \frac{7}{5}x_1 & +\frac{13}{5}x_2 & +\eta & \leq & \frac{34}{5} \quad u^1 = \left(\frac{2}{5}, \frac{1}{5}\right) \\
 & x_1 & +2x_2 & +\eta & \leq & 6 \quad u^2 = \left(\frac{1}{2}, 0\right) \\
 & 3x_1 & +6x_2 & +\eta & \leq & 18 \quad u^3 = \left(\frac{3}{2}, 0\right) \\
 & x & \leq 2, & x \in \mathbb{Z}^2_+, & \eta \in \mathbb{R}^1. &
\end{array}$$

Benders with Multiple Subproblems

In many applications, the MIP has block diagonal structure of the form

$$
\begin{array}{lllllll}
\max cx & + & h^1 y^1 & + & h^2 y^2 & + \cdots & +h^K y^K & \\
F^1 x & + & G^1 y^1 & & & & & \leq d^1 \\
F^2 x & + & & & G^2 y^2 & & & \leq d^2 \\
 & \ddots & & & & \ddots & & \leq \vdots \\
F^K x & + & & & & & G^K y^K & \leq d^K \\
 & x & \in X, & & y^k & \in \mathbb{R}_+^{p_k} & \text{for } k = 1, \ldots, K &
\end{array}
$$